

AIOps Training

Week 1: Understanding IT Operations and AIOps Basics ; Focus: The "Why" and "What" of modern operations.

Day 1: What is IT Operations? : Understanding system monitoring, incident management, and SLAs. >> Theory with example

Day 2: What is AIOps? : Goals: Observability, Automation, and Intelligence. : AIOps vs. Traditional Monitoring. >> Theory with example

Day 3: Real-World Examples of AIOps : Case studies: Predictive maintenance, noise reduction in alerts. >> practical example

Day 4: Types of Data in IT Ops : Logs (unstructured), Metrics (structured), Traces (request flow). >> Theory with example

Day 5: Setting Up Python Environment : Install Python, Python IDE (VS code), Github repo access , Jupyter Notebook , Create Project Structure environments, splunk etc. >> hands On Coding

Week 2: Simulating and Working with Logs and Metrics ; Focus: Getting your hands on the real ITSM data.

Day 6: Simulating Log Files : Generate sample log files (Apache/Nginx style) using LLM. >> log hub data , >> hands On Coding

Day 7: Reading and Parsing Logs : Use built-in file handling and Regex to extract IPs, timestamps, and errors, infos. >> hands On Coding

Day 8: Generating System Metrics : Use psutil library to monitor CPU, memory, and disk usage in real-time. >> hands On Coding

Day 9: Storing Logs and Metrics Locally : Saving parsed data into CSV and JSON formats and pandas data frame for analysis . >> hands On Coding

Day 10: Visualizing DataUse matplotlib and seaborn to plot CPU trends and error spikes. >> hands On Coding

Week 3: Introduction to Machine Learning Concepts ; Focus: The "AI" in AIOps.

Day 11: What is Machine Learning? ; Supervised (Classification) vs. Unsupervised (Anomaly Detection) learning. >> Theory with example

Day 12: Using scikit-learn for ML : Simple classification and clustering examples. >> hands On Coding

Day 13: Feature Extraction from Logs : Bag-of-words and tokenization with CountVectorizer, Embedding model and DBScan from sklearn. >> hands On Coding

Day 14: Detecting Anomalies in Metrics : Statistical methods: Z-score, IQR (Interquartile Range) method. >> hands On Coding

Day 15: Hands-On: Log Classification Lab : Train a model to classify synthetic logs into Info, Warning, or Error. >> hands On Coding

Week 4: Building traditional AIOps Functions with Python ; Focus: Core logic for correlation and analysis.

Day 16: Event Correlation Simulation : Group events by timestamps and hostnames using pandas. >> hands On Coding

Day 17: Root Cause Analysis - Basic Approach , Identify frequent patterns , Agentic Investigate feature of GCP. >> hands On Coding, demonstration

Day 18: Visual Alerting :Use matplotlib to highlight threshold breaches (alerts) on graphs. >> hands On Coding

Day 19: Noise Reduction Techniques & Alerting System via Email : Deduplicate alerts using Python sets and time windows / splunk email alert . >> hands On Coding & demonstration

Day 20: Lab: Analyze a Log Dataset End-to-End Ingest raw logs → Parse → Classify → Visualize errors. >> hands On Coding

Week 5: Automating Basic AIOps Workflows Traditional way ; Focus: moving from analysis to action in traditional way.

Day 21: Scheduling Scripts with Python : Use schedule and time libraries to run monitors periodically.

Day 22: Log Rotation and Management : Archiving and compressing old logs using shutil.

Day 23: Alerting System via Email : Use smtplib to send email notifications when critical errors occur / splunk email alert

Day 24: Generate Incident Reports : Export analysis findings to PDF or HTML using Python (Jinja2/FPDF).

Day 25: Lab: Trigger Alert if CPU > 80% : Build a script that monitors live CPU and emails you if the threshold is crossed.

Week 6: Agentic AI with MCP in AIOps

Day 26: Investigation mode using agents >> hands On Coding

Day 27: Impact assessment agent, understanding the downstream impact >> hands on coding

Day 28: Connect agents with MCP to connect to different ITSM tools >> hands on coding

Day 29: ITSM dynamic dashboard with agentic AI >> hands on coding

Day 30: Lab: How to monitor Agents in future (AgentOps in AIOps) >> Theory with example

Week 7: Final Capstone

Day 31: Building a RCA Agent which can connect to multiple systems , hypothesize the possibilities, identify the root cause - phase 1 >> hands On Coding

Day 32: Building a RCA Agent which can connect to multiple systems , hypothesize the possibilities, identify the root cause - phase 2>> hands on coding

Day 33: Proactive agent to detect the pattern and trigger alert : in traditional way - phase 1 >> hands on coding

Day 34:Proactive agent to detect the pattern and trigger alert : in agentic way- phase 1 >> hands on coding

- Day 35: Outro and overflow >> Theory with example